Dohmatob et al. [16] theorize that increasing model and/or dataset size amplifies model collapse once data quality becomes sufficiently low. Building on this theory, we can assume that low-diversity ecosystems ($D \leq 2$) tend to operate in a low data-quality regime, leading to stronger collapse as model size increases, while the opposite holds for high-diversity ecosystems ($D \geq 4$). This raises the question of how diversity helps preserve high data quality, given that diverse ecosystems start with lower model performance at iteration step $t = 0$. To address this, we introduce *effective data quality (EDQ)*, which builds on the framework of Dohmatob et al. [16]. We then use this hypothesis to explain our empirical results in detail.

Effective data quality EDQ

Training data quality reflects the discrepancy between $\hat{p}data$ and $pdata$ [16, 65], for example as measured by [8, 56]: $\delta(\hat{p}_{data}) = \mathbb{W}_2(\hat{p}_{data}, p_{data})$, whereas $\mathbb{W}_2$ refers to the Wasserstein-2 distance between the two distributions[6]. For an original (non–model-generated) dataset, all examples lie within the support of the true distribution, and this discrepancy approaches zero as dataset size approaches infinity. In contrast, due to models' internal errors (functional expressivity and functional approximation errors [56]), resampled artificial datasets can contain less information about the true distribution and can even include examples outside its support. Moreover, depending on the data received in previous training steps, not every in-distribution example provides new or useful information to a model [22, 58] and further training on a dataset similar to the initial training set risks performance decay via overfitting [61]. Effective fine-tuning methods suggest that optimal training data is not merely data that reflects the true distribution, but data that most efficiently moves model weights toward that distribution [5, 14, 35, 42, 48]. Hence, optimal data must account for the model's existing distribution, which approximates a mixture of previous training distributions, i.e., $p_{\theta(t)} \approx \sum_{i=0}^{t-1} \hat{p}_{data(i)}$. We define EDQ at iteration t as the difference between the training–truth discrepancy and the model–truth discrepancy[7]:

$$\delta_{EDQ}(\hat{p}_{data}, p_\theta) := \mathbb{W}_2(\hat{p}_{data(t)}, p_{data}) - \mathbb{W}_2(p_{\theta(t)}, p_{data}), \tag{7}$$

where lower values of δ_{EDQ} indicate higher EDQ. EDQ reflects the model-dependent informativeness of the given training data. Using δ_{EDQ}, we can now state our main hypothesis under finite training data and compute:

Hypothesis: EDQ, informal

(1) **High-EDQ regime:** If $\mathbb{W}_2(\hat{p}_{data(t)}, p_{data}))$ sufficiently lower than $\mathbb{W}_2(p_{\theta(t)}, p_{data})$, additional training improves performance.
(2) **Low-EDQ regime :** If $\mathbb{W}_2(\hat{p}_{data(t)}, p_{data}))$ sufficiently greater than $\mathbb{W}_2(p_{\theta(t)}, p_{data})$, additional training degrades performance.
(3) **Medium-EDQ regime:** If $\mathbb{W}_2(\hat{p}_{data(t)}, p_{data})) \approx \mathbb{W}_2(p_{\theta(t)}, p_{data})$, additional training neither significantly improves nor degrades model performance.

This hypothesis has two implications for recursively trained models. First, performance gains or losses depend on EDQ, regardless of whether the data is generated by humans or models [22, 65], or on its diversity per se. Nevertheless, both the proportion of artificial data and data diversity can shape the EDQ. Second, when operating in a low-EDQ regime, neural scaling laws invert: performance decay increases with system scale, consistent with prior findings [16, 31, 68]. Neural

[6]The probability distributions are illustrated in Figure 11 in the appendix.
[7]Because we introduce EDQ primarily to explain our empirical results, we rely on a simple scalar metric to capture distributional discrepancy and ignore additional factors required for optimal model weights updates, such as the direction of the distributional shift.

Scaling laws state that test loss depends on model size and dataset size, following a power-law relationship [33, 37].

Diversity and EDQ

The EDQ hypothesis helps explain the impact of ecosystem diversity on model performance. First, in our segmentation approach, sample size per model decreases as diversity increases; thus, once models operate in a low-EDQ regime, greater diversity becomes beneficial. This relationship is consistent with prior work [16] and reflects a trade-off in long-term considerations of AI ecosystems between large-scale under high-EDQ and small-scale under low-EDQ. Second, the distribution of an artificial dataset closely resembles the distribution of the model from which it is sampled. If a model had zero sampling error, these distributions would be identical. However, under our hypothesis, data whose distribution closely mirrors the model itself has low EDQ. In contrast, in a diverse ecosystem, different models represent different components of the true distribution. As a result, the EDQ of redistributed outputs can be substantially higher than in a setting where models only see their own output: data that is not useful for the model that generated it may still be useful for another model within the same ecosystem (see Illustration 12 in the Appendix). This introduces a second trade-off for maximizing EDQ: between larger datasets per model when the data are fresh to the model ($t = 0$) and higher EDQ through diversity in later iterations.

We now use the proposed theory to interpret performance dynamics across different stages of self-training iterations.

Iteration step $t = 0$ At $t = 0$, all models are trained on subsets of real data and therefore operate in a high-EDQ regime. EDQ increases with sample size per model $n = N/M$, which decreases as the number of models M increases. Because default scaling laws apply in this regime, performance decreases with diversity and improves with system scale (see Section 4.1).

Later iterations In later iterations, more diverse ecosystems can exhibit lower performance loss (at $t = 1$) and/or greater performance gains because (i) the exchange of artificial data among models increases EDQ, and (ii) once models enter the low-EDQ regime, larger per-model sample sizes n become harmful. This explains the sharp decay observed for $D = 1$ in Figure 4, which is further amplified by larger models (Figure 5). Performance trajectories after $t = 0$ illustrate that models in diverse ecosystems receive data of higher EDQ than models in homogeneous ecosystems, ultimately yielding superior long-term performance, despite lower model performance at the beginning.

Post hoc sensitivity analyses of individual factors that may influence the performance dynamics observed in the evolution of AI ecosystems support our EDQ hypothesis (see Appendix H). The results indicate that neither (model-independent) data quality, training data size, the number of models per ecosystem, nor overfitting can fully account for the observed performance decay or the differences across ecosystems.

6 Discussion

In our study of AI ecosystem evolution, we find that the optimal level of diversity increases monotonically with the number of self-training iterations (see Figure 3).

This offers a new perspective on generative model performance. Training a single model on as much data as possible is short-sighted: over longer horizons, community models trained on segments of the available data outperform a single model with access to the full dataset. The more often artificial data is recycled for training, the greater the degree of segmentation needed to attain the optimal diversity level. Put simply, the benefits of diversity grow with the number of iterations.

This observed effect is robust across model families, model sizes, dataset scales, and experimental variations involving real data and temperature-based sampling. Moreover, we find that the benefits of diversity also grow as both models and datasets scale. Taken together, our results suggest that

differences between models are a key mechanism for mitigating collapse even under more realistic conditions where models are larger, trained on far more data, can be optimized for diverse outputs, and where human-generated data accumulates alongside model-generated data.

Before turning to sociotechnical implications for AI evaluation and governance, we briefly restate the technical aspects from Section 5. To interpret our empirical results, we introduce *effective data quality* (EDQ), which treats the informativeness of data as a function of both the model distribution and the true distribution[8]. When EDQ is sufficiently high, additional training improves performance; when EDQ is low, further training is harmful and performance declines. This framing has two broad implications for recursively trained models and machine learning more generally. First, gains and losses depend on EDQ regardless of whether the data is human- or model-generated [5, 22, 65]. Second, in low-EDQ regimes, standard scaling laws invert: performance degradation increases with scale, consistent with prior findings [16, 31, 68]. Through the lens of EDQ, diversity helps prevent collapse for two reasons: (i) once the low-EDQ regime is reached, reducing data size per model via segmentation becomes beneficial; and (ii) distributing model-generated data across diverse models increases the model-dependent informativeness of the data each model receives. By contrast, increasing data per model (and reducing diversity) is only beneficial early on, when models are first exposed to new data and EDQ is dominated by sample size. The advantage of monoculture observed short-term can even disappear entirely under three modeling variations: introducing diversity by combining model families rather than segmenting data, instantiating diversity via dataset expansion rather than dataset segmentation, and evaluating community models on community-specific test sets rather than a universal benchmark. These findings, reported in the Appendix G, suggest that multiple forms of model diversity potentially can potentially be combined to mitigate collapse.

How should these results inform AI goals and governance? Most directly, the long-term epistemic risks of AI monoculture challenge the premise of a single universally optimal model, aligning with prior calls for diversity motivated by justice, fairness, and safety [9, 23, 27, 34, 40, 44]. Importantly, promoting diversity in AI does not imply endorsing inferior models; rather, it acknowledges that there are multiple legitimate conceptions of what constitutes beneficial AI.

Concerns about AI-driven knowledge collapse [49, 68] reflect the need of evaluation frameworks that capture long-term ecosystem effects—effects that short-term benchmark performance may miss. Our results suggest that model similarity (i.e., ecosystem homogeneity) can serve as an indicator of collapse risk. We welcome recent efforts to measure epistemic diversity both *within* [68] and *among* [71] models, and encourage AI operators to assess their models' disagreement relative to other models deployed in the same ecosystem.

Policy approaches from other domains may offer useful analogies, including biodiversity protection, DEI initiatives in science, and regulations that limit excessive genetic homogeneity (e.g., cousin-marriage laws). In AI, governance could reward model outputs increasing diversity in shared online data pools or regulate the distribution of critical resources such as training data. Diversity can also be incentivized bottom-up through model specificity: requiring systems to be explicit about the tasks, communities, languages, values, etc. they serve [9] would naturally push ecosystems toward greater differentiation. As long as models remain beneficial within their specified contexts, too much diversity should not be a concern.

Given the current trajectory toward AI monoculture [40, 59, 71], a shift toward greater model diversity appears necessary to sustain long-term knowledge production.

[8]Importantly, even if the EDQ hypothesis were falsified, the empirically observed diversity optimum would remain.